1. Determinare le soluzioni dell'equazione $\sin\left(x - \frac{\pi}{2}\right) = \cos(2x)$ appartenenti all'intervallo $[0, 2\pi)$.

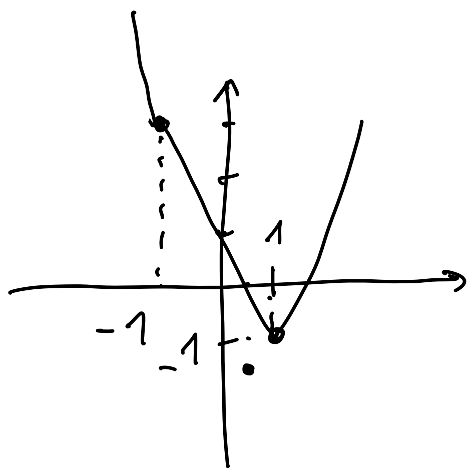
2. Risolvere la disequazione $|3^x + 3| \leq 5$.

3. Risolvere la disequazione $\dfrac{1}{2\sqrt{x}} < \dfrac{\sqrt{x}}{\sqrt[3]{x}}$.

4. Risolvere l'equazione $2(\log x)^3 - \log x = \log x^2$. [Col simbolo log si intende il logaritmo naturale, in base e.]

5. Tracciare il grafico di $f(x) = |1 - x^2| + x^2 - 2x$.

ITINERE 1 - ORE 10.30

1	$x = \pi \qquad x = \frac{\pi}{3} \qquad x = \frac{5}{3}\pi$
2	$x \le \log_3 2$
3	$x > \frac{1}{2\sqrt{2}}$
4	$x = 1 \qquad x = e^{\sqrt{3}/2} \qquad x = e^{-\sqrt{3}/2}$
5	

1)

$$\sin\left(x-\frac{\pi}{2}\right) = \cos 2x$$

$$-\cos x = \cos^2 x - \sin^2 x$$

$$\cos^2 x - (1-\cos^2 x) = -\cos x$$

$$2\cos^2 x + \cos x - 1 = 0$$

$$\cos x = \frac{-1 \pm \sqrt{1+8}}{4} = \begin{cases} -1 \\ 1/2 \end{cases}$$

$x = \pi \qquad x = \dfrac{\pi}{3} \qquad x = \dfrac{5\pi}{3}$

2)

$$|3^x + 3| \leq 5$$

$$\begin{cases} 3^x + 3 \geq 0 \\ 3^x + 3 \leq 5 \end{cases}$$

$$\begin{cases} 3^x + 3 < 0 \\ \text{mai verificata} \end{cases}$$

$$3^x \leq 2$$

$$x \leq \log_3 2$$

3) $$\frac{1}{2\sqrt{x}} < \frac{\sqrt{x}}{\sqrt[3]{x}}$$

$x > 0$

$$\sqrt[3]{x} < 2x$$

$$2x - \sqrt[3]{x} > 0$$

$$\underbrace{\sqrt[3]{x}}_{>0}\left[2\sqrt[3]{x^2} - 1\right] > 0$$

$$2\sqrt[3]{x^2} - 1 > 0$$

$$\sqrt[3]{x^2} > \frac{1}{2}$$

$x > 0$

$x^2 > \frac{1}{8}$ $\qquad$ $x > \frac{1}{2\sqrt{2}}$

4)

$$2(\log x)^3 - \log x = \log x^2$$

$$2(\log x)^3 - \log x = 2\log x$$

$x > 0$

$$2(\log x)^3 - 3\log x = 0$$

$$\log x\,[2\log^2 x - 3] = 0$$

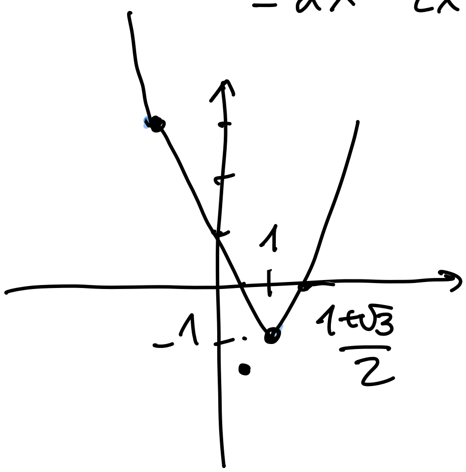
$\log x = 0$ $\quad$ $\log x = \sqrt{\frac{3}{2}}$ $\quad$ $\log x = -\sqrt{\frac{3}{2}}$

$x = 1$ $\quad$ $x = e^{\sqrt{3/2}}$ $\quad$ $x = e^{-\sqrt{3/2}}$

5)

$$f(x) = |1-x^2| + x^2 - 2x$$

$$= \begin{cases} 1-2x & -1 \le x \le 1 \\ -1+2x^2-2x \\ = 2x^2-2x-1 & x<-1 \text{ e } x>1 \end{cases}$$



$$x_v = \frac{1}{2}$$

$$y_v = \cancel{2}\cdot\frac{1}{\cancel{4}_2} - \cancel{2}\cdot\frac{1}{\cancel{2}} - 1$$

$$= -2+\frac{1}{2} = -\frac{3}{2}$$

$$2x^2-2x-1=0$$

$$x = \frac{1\pm\sqrt{1+2}}{2} = \frac{1\pm\sqrt{3}}{2}$$